Atmospheric Belonging

Diffracting Participatory Design within Troposphere

Xinquan Wen
School of Arts and Communication
Malmö University
Malmö, Sweden
xinquan.wen@mau.se

Abstract

This doctoral research explores how PD (Participatory Design) can foster "atmospheric belonging" by inviting people to collectively engage with the sky, clouds, and weather as part of climate conversations and experiences. Grounded in feminist diffraction concept and the notion of belonging, the study investigates cloudwalk – an approach initiated by the author that blend scientific knowledge and creative experience to support inclusion and multispecies participation. Key challenges addressed include balancing reflection-diffraction and science-metaphor when diffracting PD, coherence with site-specificity when practicing cloudwalk and engaging both immediate and long-term transformations in landscape and climate. The work advances diffractive participatory design methods and contributes to conference themes by illustrating how attention to hydrological cycles can encourage dialogue, equity, and coexistence among human and more-than-human communities.

CCS Concepts

• **Human-centered computing** → Interaction design; Interaction design process and methods; Participatory design.

Keywords

diffraction, more-than-human, belonging, cloudwalk, human-world relationships

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1 Overview of the Doctoral Project

1.1 Motivations and Research Question

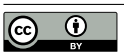
My doctoral project inquires how design can help people feel belonging with their urban environment by paying attention to the sky and the changing climate. The motivation behind this work comes from seeing how urgently climate change affects all of us but also noticing how disconnected and powerless some people feel about what is happening [11]. I want to understand how participatory design can encourage more people to join conversations and actions about climate, especially those whose voices aren't always heard.

I focus on the idea of "atmospheric belonging" - how people relate to the sky, clouds, and weather as part of their home and everyday life. I am interested in making space for people and more-than-human entities to be involved in design and decision-making. My research questions revolve around how design can deepen this sense of belonging, and how including entities like clouds and weather systems can enrich participatory processes. My evolving research question is: How to foster deeper human-world relationships through diffractive participatory design approaches?

1.2 Theoretical Approaches - Diffraction, Vibrant Matter, and Belonging

Theoretically, my work is grounded in diffraction, a concept from Donna Haraway [13] and Karen Barad [3] that helps me understand how differences and entanglements create new patterns of knowledge and relationship, rather than simple divisions or opposition. Diffraction emphasises uncertainty, multidirectionality, and the creative value of difference, showing how diverse perspectives and agencies work together to produce hybrid, artful forms of knowledge. This approach enables transformative design research that attends to emergence rather than fixed categories. In my cloudwalk practice, diffraction helps me balance scientific knowing and embodied experience, weaving together cloud classifications and atmospheric science with practices like breathing collectively and sensing movement together. Through this, participants experience "cloudtime" [16] - the living rhythms of clouds that move beyond human schedules and invite attention to ongoing, unpredictable patterns shaped by atmospheric elements, including wind, humidity, and sunlight.

Bennett [5] introduces the concept of "vital materiality," arguing that agency is distributed across assemblages of both human and nonhuman entities, rather than being exclusive to people. Her work dissolves the dichotomy between subject and object, describing how everyday things - such as trash, worms, metal, and even electricity - can act as "actants," shaping outcomes and influencing events in the world. By recognising "thing-power," Bennett suggests that we might develop a politics less focused on assigning blame and more attuned to the complex webs of causality in which both humans and things are engaged. Vital materiality supports diffracting PD through further strengthening the avoidance of predefined object-subject, such as inert matter and empowered humans.



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Besides diffraction and vibrant matter, I draw on the framework of "belonging without othering," developed by john a. powell and Stephen Menendian [17], which provides a way of thinking about inclusion that resists creating outsiders. Rather than framing belonging as fitting into fixed categories, this approach calls for relational spaces that honour difference without marginalisation. Combined with attention to intersectionality [7], which reminds us that identities are layered and dynamic - shaped by markers like language, migration status, and age - these concepts challenge participatory design to move beyond narrow notions of representation towards spaces of inclusion that acknowledge complexity without reducing individuals to singular identities.

1.3 Methodological approaches – Design Research, Diffraction, Walking

My study adopts an empirical approach, combining design research and diffraction in a methodological framework. Design research enables learning through making, ensuring that knowledge remains situated, iterative, and evolving rather than fixed. In particular, I follow research through design [15, 18], engaging directly in the process of designing and creating artefacts or interventions to explore research questions.

In my research, diffraction functions as both a theoretical lens and a methodological approach. Conceptually, diffraction focuses on how differences are relational rather than fixed or predetermined; it does not represent differences as inherent essences or consequences but attends to the patterns created by these differences. Methodologically, diffraction refrains from predefined distinctions between subject and object, instead allowing these boundaries to emerge through the research process itself.

Unlike traditional interdisciplinarity, which often reinforces disciplinary silos, diffraction allows multiple perspectives to interact without merging or subordinating one another [12]. Barad conceptualises diffraction as a means of recognising how the world takes shape differently through various practices, emphasising that knowledge does not exist as a fixed external truth but instead emerges through intra-actions, continually shaped by observation, materiality, and environmental entanglements. This perspective is particularly important for rethinking PD through a pluriversal lens, ensuring that knowledge production remains entangled with ethical, material, and ecological considerations.

In developing my empirical strategies, I engage with walking methodologies. The diverse histories of walking in artistic practice reveal walking's power to transform spaces, as walking can culturally alter the meaning - and thus the very nature - of a space without leaving material traces [6]. Drawing inspiration from contemporary artists like Francis Alÿs, I seek to foster layered meanings and invite political engagement through my research activities. Recognising the biological, social, and universal dimensions of walking, I join other artistic researchers [9, 20] in aiming to maintain its multidimensional characters - encompassing political, ethical, embodied, and liminal aspects [4, 8, 14, 19].

1.4 Empirical Approaches - Cloudwalk

My main empirical approach is cloudwalk, which invites people to spend time together observing, sharing, and imagining their relationships with the atmosphere. Each cloudwalk encourages participants to reflect on scientific facts, personal stories, and embodied experiences, all woven together through simple activities like walking, breathing, and creative artmaking. The walks have happened in city parks, nature reserves, and playgrounds. For example, together with MoB (Museums of Belgium), I facilitated six collaborative cloudwalks with the kiterunners – the youth volunteers of the organization - before and after the kite-making events, where teenagers connected artistic activities to conversations about clouds and local climate.

Clouds embody a vibrancy that resists fixed boundaries - visible entities unfolding and dissolving in flux, they are simultaneously material forms and atmospheric forces. Their scales and transience stretch the limits of human apprehension: on a cloudy day, hundreds of thousands of tons of water drift above, carrying the weight of a town. Yet most transformations at this scale escape immediate perception; they unfold over geological time, requiring the patience of timelapse to render their slow metamorphoses visible. Viewed in this light, clouds offer a lens for participatory design that is as delicate and situated as the apparatus of a quantum experiment, where specificity is always the product of entanglement and becoming.

Rather than suggesting the fixed stability of materiality, Bennett invites us to conceive of matter as inherently dynamic and alive: "my goal is to theorise a materiality that is as much force as entity, as much energy as matter, as much intensity as extension" [5]. From such a perspective, clouds stand as living exemplars of vibrant matter - challenging us to attune to both their fleeting presence and their ongoing, energetic transformation.

1.5 Progress and Plans

Progress so far includes organising and leading multiple cloudwalks, building partnerships with youth groups and educators, and developing ethical guidelines - especially important when working with minors or marginalised people. I write about these experiences and keep refining my approach. I have shared early findings at seminars and conferences, and regularly reflect with supervisors and peers, adapting and growing the project along the way. For the next stages, I plan to continue community collaborations in Malmö and Brussels, expand the approach, and deepen the theory around cloudtime and atmospheric belonging, making sure the work is accessible to different communities.

2 Main Challenges in the Ph.D. Journey

2.1 How to balance reflection-diffraction and science-metaphor when diffracting PD?

In order to introduce the challenges I am facing when diffracting PD during my doctoral research, I need to unfold the discussions through reviewing two papers on diffraction.

Through extensive literature review, Udén [21] pointed out the shortcomings of using diffraction as an interdisciplinary metaphor, such as disconnecting from its physical roots - losing scientific rigour - and simplifying diffraction to the opposition to reflection. It is an insightful reminder that a new reflection-diffraction binary could be formed when practising diffraction. Moreover, when we refer to diffraction, it would be helpful to clarify whether we utilise the notion as a scientific concept or as a metaphor. I do see the

value of both, diffraction as a scientific concept is not necessarily better than as a metaphor, which I will discuss in the next paragraphs. However, it is beneficial to be precise for the purposes of communications. In addition, regarding the emergence of the diffraction-reflection polarity, Udén reminded us that reflection is such a fundamental phenomena in our everyday life, it is questionable whether we could possibly stand on the opposite side of reflection. Just as Alendar's quote [1] in Udén's paper "It is worth pointing out that the diffraction pattern we see is a reflection." (translated by Udén)

While Udén examines diffraction as a novel feminist concept from a distance, Avila [2] adopts a diffractive approach and cross-read Félix Guattari's The Three Ecologies [10], Karen Barad's Meeting the Universe halfway [3], and the concept of intersectionality coined by Kimberle Crenshaw [7]. Avila's discussions on the resonances between these theories were insightful. For instance, "One could say that she 'diffracted' the 'social register' of feminist thinkers of that time to render 'visible' power dynamic of affecting some human groups". Through bridging intersectionality and diffraction, Avila highlights the boundary/section-crossing nature of both and how new confluences 'greater than the parts'. Moreover, through borrowing a latter concept to elaborate a previous one, Avila diffracts the sections of time - past and future - in his own writings. Hence, we could "counteract binaries such as natural-artificial and allow us instead to conceive the continuum through differences of degree and intensity."

However, I feel the case study within Avila's paper - focusing on one plant (Passiflora caerulea) and its main pollinator (Xylocopa Artifex) - was too isolated and solely science based. While Udén criticising the epidemic of metaphorising diffraction, I consider a solely science-based approach to diffraction can also lose its essence. As design researchers, we hold accountabilities to delve into the incalculable nature of natural creativity [5]. Without questioning and tapping into the unknown, incalculables, and eco-social entanglements, design researchers can be instrumentalised to serve the scientific findings, despite their validity and relevance. When we invite ourselves to explore the unknown and incalculables, the importance of utilising diffraction as metaphor could manifest itself.

In light of the previous discussions, when diffracting PD, my main challenges are how to balance reflection-diffraction and science-metaphor when diffracting PD? How to attend to patterns of difference rather than difference in PD? How to avoid flattening out when practising diffraction?

2.2 How can Cloudwalk find the balance between a coherent practice and a contextual approach?

During my fieldwork, one of the most significant challenges I face is learning how to make cloudwalks properly contextualised and site-specific. Earlier iterations of my cloudwalks focused primarily on the sky, grounded in the recognition that clouds are bodies of water - floating, ever-shifting entities borne by wind. These walks were highly adaptable, unfolding wherever the sky offered clouds, but their flexibility often meant they lacked contextual or site-specific grounding.

I unfold the discussions regarding the challenges of contextualising cloudwalk through reviewing one cloudwalk held in Kiewit, Belgium. The Kiewit cloudwalk, conducted in September 2025, was my first example of attempting contextual adaptation, where I tried to attune to the transformations of the local landscape and the relevant shifts in hydrological cycles.

In Kiewit, I researched how the area was composed mostly of open heathland and pastures until the twentieth century, with only subtle traces of its pre-residential era remaining in street names like Rietstraat and Waterleliestraat (meaning reedstreet and waterlilystreet in English). At a stop during the walk in De Wijers nature reserve, reeds still persisted as living memory of the area's past. However, I realise now that whilst I approached what Donna Haraway calls "culturenature" - engaging with cultural transformations like the annual Pukkelpop festival that completely transforms the space - I was less attentive to "natureculture." I overlooked that even nature reserves are culturally constructed concepts, managed and regulated, fragmented and divided by private properties. The participants had individual experiences at De Wijers that I failed to draw out, assuming that visits to nature reserves provide universal experiences whilst festivals trigger more personal responses.

Another temporal challenge has emerged around the scales of transformation I attend to. I attuned to the more obvious annual transformations of landscapes - like the seasonal transition from summer to autumn, or the dramatic yearly change brought by music festivals - whilst I didn't dive deep enough into the transformations within decades or centuries timeframes. This represents a significant gap in my approach to natureculture and culturenature, as longer historical changes shape hydrological cycles and atmospheric patterns in ways that annual cycles cannot capture. Understanding how to weave together these different temporal scales - from cloudtime to seasonal time to historical time - remains an ongoing methodological challenge.

Additionally, I overlooked the site specificity of clouds themselves. Whilst I introduced cloud types like cumulus and stratocumulus during the Kiewit walk, I realise I could have connected more meaningfully to the hydrological theme by diving deeper into regional climate and atmospheric patterns typical of the North Sea and mid-latitude Western Europe. Each place has its own atmospheric character shaped by geography, seasons, and human activity, and learning to read and respond to these specificities is crucial for making cloudwalks truly contextual rather than generic sky-watching exercises.

Through Kiewit Cloudwalk example, when practising Cloudwalks, my main challenges are how can Cloudwalk find the balance between a coherent practice and a contextual approach? How can Cloudwalk be truly site-specific, attuning to glocal hydrological cycles both on the land and in the sky? How to weave together these different temporal scales - from cloud time to seasonal time to historical time?

3 Intended Contributions to Participatory Design and Conference Theme

This research contributes to participatory design by creating new ways for people to experience civic participation and address climate crisis together. Diffractive approaches I have been developing



Figure 1: Cloudwalk Kiewit. (photo by Youshan Li, anonymised by author, used with permission).

move beyond problem-solution frameworks to embrace complexity, uncertainty, and emergence, offering new theoretical and practical tools for design researchers working with environmental challenges.

3.1 Civic participation and climate crisis

Regarding civic participation and climate crisis, cloudwalks provide embodied, place-based ways for people to come together in public conversation about atmospheric changes. Through these practices, participants learn to attune to hydrological cycles - recognising clouds as bodies of water that were once forests, lakes, or oceans, and will become future rivers, grass, or rain. This attunement helps people understand their local environment as part of a living, global water system, making abstract climate processes tangible and personally relevant. Cloudtime invites participants to step out of human schedules and experience temporality from the clouds' perspective, shaped by wind, humidity, and sunlight. This temporal awareness provides alternative, informal ways to learn about climate science whilst cultivating collective awareness that goes beyond individual action.

3.2 Diffracting PD

The diffractive approach to participatory design that I am developing offers different possibilities for civic engagement by embracing uncertainty and multidirectionality rather than linear problem-solving.

Diffracting PD can provide at least the following three contributions. Firstly, through cross-reading, boundary-crossing, and forming new confluences, PD practitioners could connect to seemingly unrelated yet potentially relevant fields when diffracting PD. For instance, within cloudwalks, cloud-observing, body relaxation, and drinking water are linked and in dialogue to each other. Secondly, referring to diffraction as a scientific concept or metaphor according to situations, could allows PD practitioners navigate through scientific findings and have the courage to tap into the incalculables and unknown. For instance, during cloudwalks, scientific knowledge about clouds, such as hydrological cycles are shared and discussed. At meantime, we talked about the incalculable, such as the shift between landscape and skyscape through transpiration, evaporation, and precipitation. A realm we need to enter through imaginations and discussions. For example, when pieces of clouds above our sky can originated hundreds of kilometers away. Lastly, diffractive methodologies provide a safe place when subjects of research do not feel comfortable to be spotlighted on. For instance,

both the organisers and kiterunners of MoB are not comfortable to respond to questions and interests, which are solely focusing on the migrant background of the young kiterunners. A diffractive approach allows us to build connections and trust through our common interests - the sky. The conversations around the countries we previously lived just naturally occurred.

These potential contributions of diffracting PD enable design processes that are more responsive to emergence, more attentive to complexity, and more capable of working with the ongoing transformations that characterise both climate change and social change.

3.3 Inclusion and equity

Regarding inclusion and equity, my approach recognises participation as an ongoing, relational process. Drawing on belonging without othering and intersectionality, cloudwalks allow marginalised voices to shape their own experiences with atmospheric environments. Creative practices like kite-making, atmospheric playlists, and group reflection create accessible entry points that honour different ways of knowing and being. By working with young people from diverse migration backgrounds at MoB, the project demonstrates how participants can become stewards of local microclimates, contributing to urban climate resilience through their own creative leadership.

3.4 Multispecies participation

For multispecies participation, the project brings clouds, weather, and landscapes into collective designs, opening up new ethical questions about agency, stewardship, and shared responsibility in a more-than-human world. Through attention to cloudtime and atmospheric belonging, cloudwalks help participants recognise the entanglement between land- and sky-scapes, their constant transformation, and their role in sustaining life across scales. The glocal hydrological cycle becomes a framework for understanding how local actions and global systems are interconnected, encouraging forms of collective stewardship that attend to both immediate environments and planetary processes.

Diffractive participatory design provides different possibilities for multispecies collaboration by moving beyond symbolic gestures toward genuine engagement with more-than-human temporalities and agencies. Rather than simply representing nature, the approach creates conditions where atmospheric phenomena can actively shape the direction and outcomes of participatory processes. This contributes to broader conversations about coexistence, care, and living together through climate change, offering methods that are adaptive enough to work with uncertainty whilst robust enough to generate meaningful collective action.

In summary, my project hopes to offer practical methods, thoughtful reflections, and evolving stories that advance participatory design by fostering inclusion, civic engagement, and multispecies relationships. By sharing this work, I aim to contribute to a broader conversation about how we live together - through climate change - locally and globally, supporting the conference themes of peace, dialogue, and coexistence through design practices that nurture atmospheric belonging and collective care.

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